CTA rail ridership recovery, 2019-2025

Station entries from CTA annual reports, with a 100,000-entry 2019 threshold

Loading and preparing the station data

```
# https://github.com/zixinhu23/cta-ridership-recovery
from pathlib import Path

import pandas as pd

from scripts.cta_recovery_analysis import draw_recovery

# Station totals compiled from CTA's annual ridership reports.
DATA = Path("data/processed/cta_station_recovery.csv")
FIGURE = Path("output/figures/cta_recovery.png")

def load_rides(path):
    types = {
        "station": "string",
        "line": "string",
        "entries_2019": "Int64",
        "entries_2025": "Int64",
    }
    rides = pd.read_csv(path, dtype=types)
    rides["station"] = rides["station"].str.strip()
    rides["line"] = rides["line"].astype("category")
    return rides

def recovery_table(rides, min_base=100_000):
    rides = rides.dropna(subset=["station", "line", "entries_2019", "entries_2025"])
    rides = rides[(rides["entries_2019"] >= min_base) & (rides["entries_2025"] > 0)]
    rides["recovery"] = 100 * rides["entries_2025"] / rides["entries_2019"]
    return rides.sort_values("recovery").reset_index(drop=True)
```

Output

```
Stations analyzed: 112
Median recovery: 66.7%
Slowest stations:
    station line   recovery
   Oak Park Blue  35.501092
    Pulaski Blue  35.751107
     Racine Blue  36.562495
     Austin Blue  38.941886
Forest Park Blue  40.169913
Median recovery by line:
line
Yellow    53.1
Purple    59.6
Brown     63.8
Red       64.1
Blue      64.7
Orange    69.3
Loop      69.8
Green     72.0
Pink      77.9
```

The recovery ratio uses 2019 station entries as its denominator.

Source: CTA Annual Ridership Reports, CY2019 and CY2025. Rates are shown for matched stations only.

Recovery by station and line

The five slowest stations are all on the Blue Line's Forest Park branch

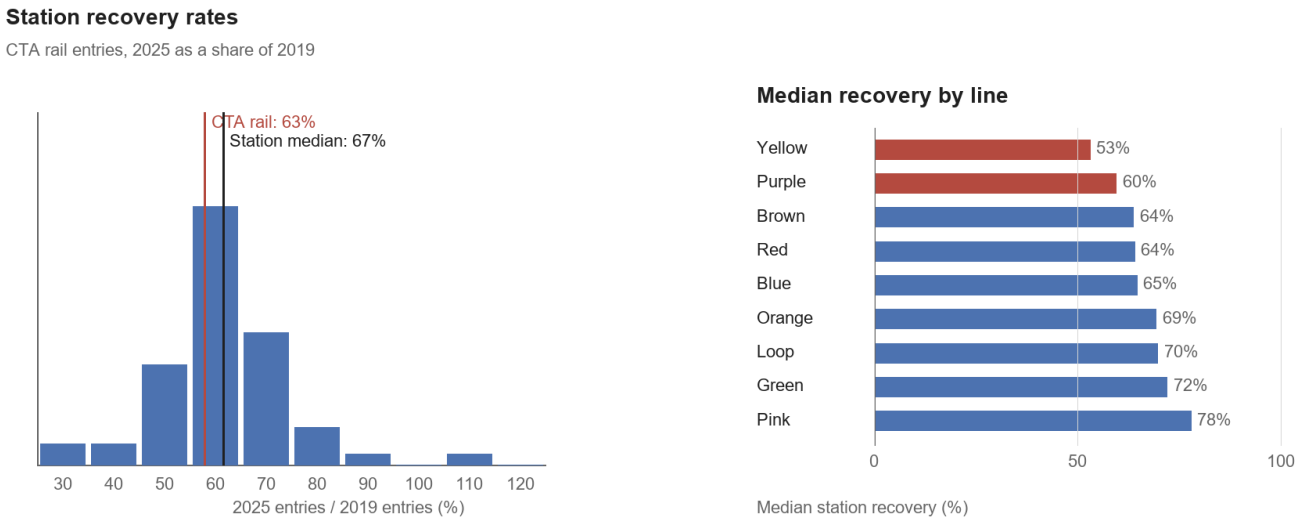


Figure 1. Station recovery rates and median recovery by line.

Summaries and figure call

```
def line_profile(stations):
    return stations.groupby("line", observed=True)["recovery"].median().sort_values()

rides = load_rides(DATA)
stations = recovery_table(rides)
line_medians = line_profile(stations)
slowest = stations.nsmallest(5, "recovery")

print(f"Stations analyzed: {len(stations)}")
print(f"Median recovery: {stations['recovery'].median():.1f}%")
print("\nSlowest stations:")
print(slowest[["station", "line", "recovery"]].to_string(index=False))
print("\nMedian recovery by line:")
print(line_medians.round(1).to_string())

draw_recovery(stations, line_medians, FIGURE)
```

Findings

Among the 112 matched stations, the median recovery was 66.7%. Line medians ranged from 53.1% on the Yellow Line to 77.9% on the Pink Line.

The five lowest station rates were Oak Park (35.5%), Pulaski (35.8%), Racine (36.6%), Austin (38.9%), and Forest Park (40.2%).

Morgan (118.2%) and Cermak-Chinatown (115.5%) exceeded their 2019 totals. Cermak on the Green Line reached 97.9%.

Station entries are not unique riders. CTA's 2025 reporting changes affect comparability; these results are descriptive.